

t'Hooft Meets a String: A Proposal on Shockwave Production by Massive Particles

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Abstract

In this note, the authors propose to extend shockwave production by photons to the string theory domain and more massive particles.

In a paper with Dray, Gerard t'Hooft gives the analysis for spacetime shockwaves (gravitational) generated as a result of the massless photon [1]. In [2] (see Chapter 4.5, page 125), Becker, Becker and Schwarz mention how in the negative sector, mass is created by raising operators of the oscillator - much like high energy states of a quantum simple harmonic oscillator are created by the raising operators [3]. In the present note we propose to integrate these two results. That is, we propose the following investigation: what happens if the massless particle of Dray-t'Hooft is subjected to the raising operator? What is the shockwave or gravitational signature generated by it as the mass moves near an event horizon? This proposal would correctly be framed in the string description of black holes [4] and their event horizons, masses, and gravitational waves [5].

References

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